Fixed local gain evaluator and peer message interface

Conservative forecast caps keep Φ on linear branches. Heuristic ranking, not an expected-cost guarantee.

Fixed forecast-slack proxy

$$\rho = I + \Sigma \text{ pipeline} - \Sigma \text{ exposed forecasts}$$
$$\rho > 0: \text{donor};\ \ \rho < 0: \text{receiver};\ \ \rho = 0: \text{inactive}$$

$$\Phi(\rho) = h \max(\rho, 0) + b \max(-\rho, 0)$$

Donor marginal cost
$$A(q) = \Phi(\rho_d - q) - \Phi(\rho_d)$$

Receiver marginal benefit
$$B(q) = \Phi(\rho_r) - \Phi(\rho_r + q)$$

$$G(q) = B(q) - A(q) - C_{trans}(q)$$
$$= (h_{donor} + b_{receiver} - c)\, q - K,\ \ q > 0$$

Caps: request ≤ forecast deficit; offer ≤ forecast surplus.
A positive-gain pair takes largest q; no interior optimum.

$q \in \{0, 0.25, 0.5, 0.75, 1\} \times \min(\text{remaining offer}, \text{remaining request})$

Local message schema (illustrative)

Header: site / peer IDs, period, round
Role: request or offer
Remaining request / offer quantity
Candidate q and local marginal values
Common ranking / deterministic tie key
Confirm or release; reservation token

Private inventory, pipeline, forecasts.

Fixed protocol invariants

- Common total ranking of positive-gain edges
- Mutual best; at most one partner / round
- Reserve / confirm prevents overpromise
- No receive-forward within the period